Phase 4.25: Rank-32 Memory Localization

Tactix development report for Andrew

27 August 2026

1 Outcome

The accepted-source Rank-32 memory-localization increment is terminal. Every completed diagnostic returned UNSAT, passed independent DRAT verification, and left zero certificate and scratch residue. No solver repair is promoted.

The fresh accepted-binary screen measured Tactix/current Kissat at 582,204/299,988 KiB peak RSS (1.940757x) and 410.828/292.637 seconds. The excess is retained inprocessing allocation rather than extra search, an OOM, or a checker failure. At the stable late boundary, glibc reports 333,394 KiB used heap, 127,957 KiB free heap, and 92,412 KiB mmap. Free-heap retention is material, but used heap plus mmap alone is 415.826 MiB, so trimming cannot be the main repair.

Late live owner	Storage
Named persistent kernel owners	267.562 MiB
Dense occurrence index, allocator-usable	92.855 MiB
Persistent policy buffers	13.307 MiB
Remaining process/metadata/smaller owners	42.102 MiB
Used heap plus mmap	415.826 MiB
Allocator-retained free heap	124.958 MiB

The largest individual named capacities are reconstruction at 86.37 MiB, clause-order aliases at 45.37 MiB, and proof indices at 35.29 MiB. The dense index requests 84.44 MiB, but glibc supplies 92.86 MiB across 834,922 allocations: size-class rounding alone adds 8.42 MiB, before allocation metadata.

2 Architectural interpretation

Current Kissat reuses its per-literal watch vectors for dense occurrences. Tactix's flat sparse-watch arena is a different hot representation and cannot serve as the dense per-literal lists, so Tactix owns a separate vector-of-vectors. That is a concrete architectural reason for roughly 93 MiB plus allocator metadata, not evidence of a leaked rule-local scratch object.

A flat dense layout could reduce vector headers, allocation count, rounding, and metadata, but its observed ceiling is only tens of MiB and the rewrite would span broad inprocessing mutation paths. It is therefore deferred until cross-instance evidence makes that risk worthwhile.

The lower-risk reconstruction reserve opportunity is also bounded. An exact non-power-of-two Substitute reserve seed later doubles to 22,641,024 words for 15,649,920 logical words. Power-of-two reservation would cap this run at 16,777,216 words, saving at most 22.37 MiB, roughly 4% of the diagnostic peak. That is useful secondary cleanup, but not a meaningful closure of the roughly 266-MiB fresh-screen gap.

3 Evidence and failures

The terminal chain is frozen under `benchmarks/runs`: jobs 1652456, 1652457, 1652461, 1652463, 1652469, 1652473, 1652474, and 1652475 cover boundary ownership, rejected BVE scratch lifetime, capacity, allocator state, persistent owners, dense capacity, policy buffers, and allocator-rounded dense storage. Every solver job requested 12 CPUs and 48,000 MB, used no `RLIMIT_AS`, node constraint, exclusivity, or throttle, and retained the 16,000-second independent checker window. Runtime is diagnostic evidence only, not a performance claim.

The final result and slot-trace hashes are, respectively, `da2ebaf2a53a...f3550b121d1a` and `751686f58cd7...b1ccab65e04e`. After removing telemetry fields, all 2,640 slot records and every Congruence/Substitute logical trace are byte-identical between successive builds.

Two scaffolds failed before the solver and are frozen non-scorable: job 1652467 omitted the manifest input-hash map, and job 1652472 copied a stale wrapper binary pin. Neither produced a solver result or residue, and neither is reused. No soundness, crash, sanitizer, model, or proof failure occurred.

4 Decision and next action

No memory change is promoted or sent through the expensive three-wave/gate/fuzz path. The measured reconstruction and dense-layout ceilings do not justify delaying the solve-loss cohort.

Ranks 15, 16, 19, 23, and 36 are the five fresh Tactix timeouts, and all five instances are SAT. They now become the immediate convergence target. Fresh bounded accepted-Tactix/current-Kissat diagnostics will compare early Lucky/Walk, phase and rephase state, mode switching, decisions, and learned clause quality. Only a shared portable owner may become a repair. Any acceptance claim still requires three independent repetitions, the complete frozen B28+C5 gate, and Claude-owned fuzz. The canceled partial Rank 1–40 refresh remains excluded from selection and inference.

The detailed frozen account is `benchmarks/reports/2026-08-27-rank32-memory-localization.md`.